A Robust Test for the Stationarity Assumption in Sequential Decision Making

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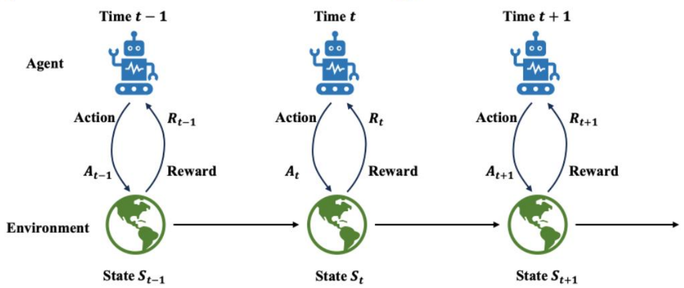
Preliminary

Sequential Decision Making



- Markov Decision Process (MDP): $p_t(S_{t+1}, R_t | A_t, S_t)$.
- Stationarity assumption

  $\boldsymbol{H_0}: \{\boldsymbol{p_j}(\boldsymbol{S_{j+1}}, \boldsymbol{R_j} | \boldsymbol{S_j}, \boldsymbol{A_j})\}_{0 \le j \le T-1}$ **are homogeneous over time** $\boldsymbol{j}$**;**
- Offline reinforcement learning (RL) : static dataset without further interactions.
- Additional challenges in modern RL problems
  - State variable S_t can be high-dimensional (e.g., 7056 for Atari).
  - Dynamics p_t can be highly complex (e.g., autonomous driving).
- Modern ML methods well-suited for non-linearity and high-dimensionality, however, need careful treatment for proper statistical inference.

- Maximum-type test statistic:

$$\hat{\Gamma} = \max_{\substack{\epsilon T \le t \le (1-\epsilon)T \\ b \in \{1,\dots,B\}}} \sqrt{\frac{t(T-t)}{T^2}}\, \hat{\sigma}(t, h_b)^{-1} \frac{1}{|I_2|} \sum_{i=1}^{|I_2|} \hat{\psi}_i,$$

where

$$\hat{\psi}_i = \underbrace{\int \left| \hat{E}_{[t,T]}[h_b(S')|a,s] - \hat{E}_{[0,t]}[h_b(S')|a,s] \right|}_{\hat{\Delta}(a,s;h_b,t)} g(a,s)\lambda(da,ds)$$

plug-in term

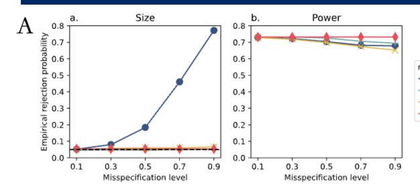
augmented term

$$+\frac{1}{T-t}\sum_{j=t}^{T-1} \mathrm{sgn}(\hat{\Delta}(A_{i,j}, S_{i,j}; h_b, t)) \left[h_b(S_{i,j+1}) - \hat{E}_{[t,T]}[h_b(S')|A_{i,j}, S_{i,j}] \right] \frac{g(A_{i,j}, S_{i,j})}{\hat{\omega}_{[t,T]}(A_{i,j}, S_{i,j})}$$

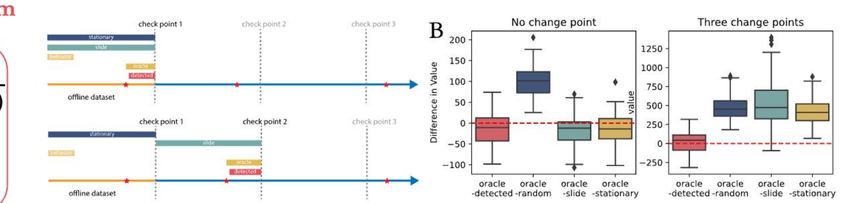
$$-\frac{1}{t}\sum_{j=0}^{t-1} \mathrm{sgn}\left(\hat{\Delta}(A_{i,j}, S_{i,j}; h_b, t)\right) \left[h_b(S_{i,j+1}) - \hat{E}_{[0,t]}[h_b(S')|A_{i,j}, S_{i,j}] \right] \frac{g(A_{i,j}, S_{i,j})}{\hat{\omega}_{[0,t]}(A_{i,j}, S_{i,j})}.$$

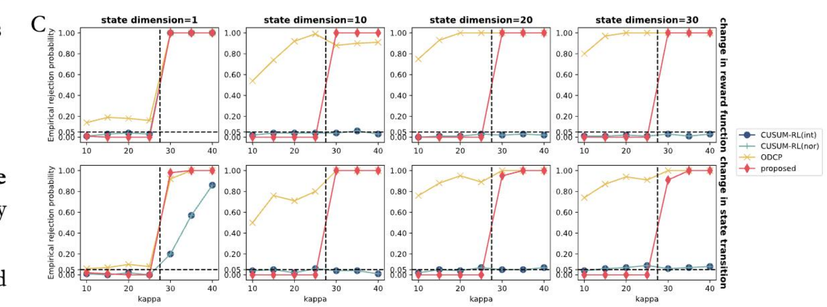
- h_b is a test function from a rich function class and B is the number of test functions; g is some reference density function. $\epsilon \in (0, 0.5)$ is the boundary removal parameter.
- $\sqrt{t(T-t)/T^2}\hat{\sigma}(t, h_b)^{-1}$ is the normalized factor for different (t, h_b) where $\hat{\sigma}^2(t, h_b)$ is the sampling variance estimator.
- Nuisance functions $\hat{p}_{[0,t]}, \hat{p}_{[t,T]}, \hat{\omega}_{[0,t]}, \hat{\omega}_{[t,T]}$ are estimated using the data from I_1. **One can use flexible machine/deep learning tools.** The estimator $\hat{\psi}_i$ is evaluated by directly plugging estimated nuisance functions and samples from I_2.
- Gaussian multiplier bootstrap is used to approximate the limiting distribution and compute the resulting p-value, i.e., $\hat{p} = \sum_{q=1}^{Q} I(\hat{\Gamma} \ge \hat{\Gamma}^q)/Q$.

Simulation



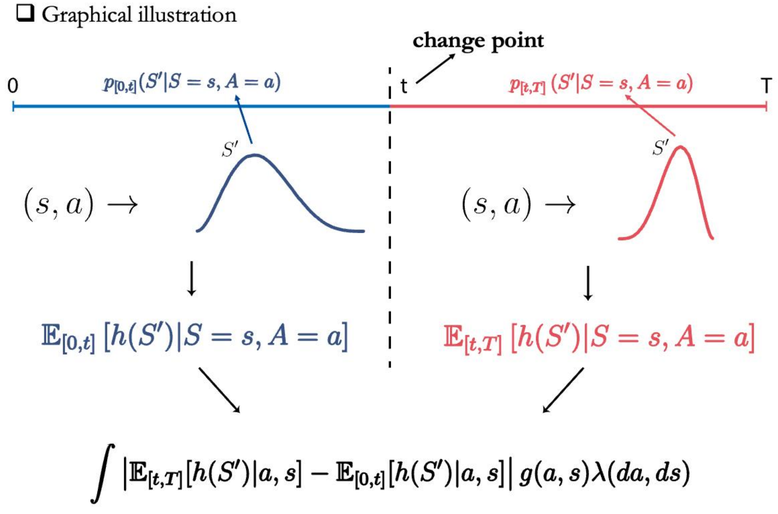
- Fig. A: Double Robustness.
- Fig. B: Applicability to online settings.
- Fig. C: Superiority under high-dimensional settings





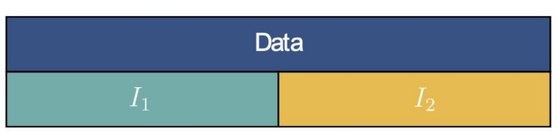
Proposed Test

- Graphical illustration



$$\int \left| \mathbb{E}_{[t,T]}[h(S')|a,s] - \mathbb{E}_{[0,t]}[h(S')|a,s] \right| g(a,s)\lambda(da,ds)$$

- The proposed test is based on **sample splitting**[1].



Theoretical Property

Double Robustness: Suppose H_0 holds. For any $h \in H$ and $t \in (0, T-1)$, we have

$$E\psi_i^* = 0, i = 1, \dots, N,$$

Moreover, the above equation is doubly robust, i.e., for any $p_{[0,t]}$, $p_{[t,T]}, \omega_{[0,t]}, \omega_{[t,T]}$, the following holds as long as either $p_{[0,t]} = p_{[t,T]} = p^*$ or $\omega_{[0,t]} = \omega^*_{[0,t]}, \omega_{[t,T]} = \omega^*_{[t,T]}$,

$$E\psi_i = 0, i = 1, \dots, N.$$

- ψ_i^* is the oracle version of $\hat{\psi}_i$ by replacing the estimated nuisance functions with the oracle ones. ψ_i is similar to $\hat{\psi}_i$ by replacing estimated ones with general functions.

Size: Suppose $B = O((NT)^c)$ for any $c > 0$ and $t = O(T)$ for any $t \in [\epsilon T, (1-\epsilon T)]$ hold. In addition, suppose uniform boundness of $\frac{g}{\omega_{[0,t]}}, \frac{g}{\omega_{[t,T]}}$, strict stationarity and geometrical ergodicity of $\{S_t\}_{t \ge 0}$ as $T \to \infty$ and the convergence speed of $\hat{p}_{[0,t]}, \hat{p}_{[t,T]}, \hat{\omega}_{[0,t]}, \hat{\omega}_{[t,T]}$ to oracle value satisfying

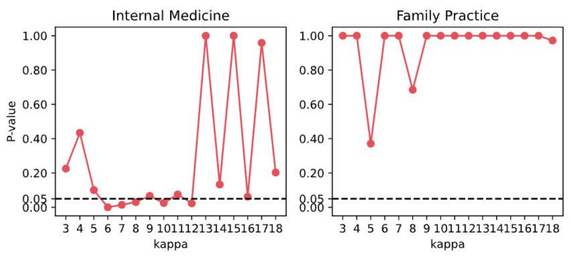
$$\left[E\{d_{TV}^2(\hat{p}_{[0,t]}, p^*_{[0,t]})\}\right]^{1/2} = O((NT)^{-\kappa_1}), \left[E\{d_{TV}^2(\hat{p}_{[t,T]}, p^*_{[t,T]})\}\right]^{1/2} = O((NT)^{-\kappa_2}),$$
$$\left[E\{d_{TV}^2(\hat{\omega}_{[0,t]}, \omega^*_{[0,t]})\}\right]^{1/2} = O((NT)^{-\kappa_3}), \left[E\{d_{TV}^2(\hat{\omega}_{[t,T]}, \omega^*_{[t,T]})\}\right]^{1/2} = O((NT)^{-\kappa_4}),$$
$$\kappa_1, \kappa_2, \kappa_3, \kappa_4 \in (0, ½) \text{ and } \kappa_1 + \kappa_3 > \frac{1}{2}, \kappa_2 + \kappa_4 > \frac{1}{2},$$

, we have

$$P(\hat{p} \ge \alpha | H_0) = \alpha + o(1)$$

Application to Intern Health Study Data

- **Intern Health Study [2]**: Wearable and survey data of 1565 medical interns in the U.S. collected for 26 weeks (from July 1, 2018, to Dec 31, 2018).
- The **goal** is to improve the health-related outcome of medical interns through mobile-based push notification.
- Every week, each intern would be either randomly assigned to notification week ($A_t = 1$) with **0.75** probability or non-notification week ($A_t = 0$).
- Intern's previous week's average daily step count, sleep duration and survey mood score are considered as the state S_t.
- We consider the next week daily average step count as the reward R_t.
- The treatment effect of push notification decrease over time.



References

[1] V. Chernozhukov et al., "Double/debiased machine learning for treatment and structural parameters," *The Econometrics Journal*, vol. 21, no. 1, pp. C1–C68.

[2] T. NeCamp et al., "Assessing real-time moderation for developing adaptive mobile health interventions for medical interns: Micro-randomized trial," Journal of Medical Internet Research, vol. 22, no. 3, 2020.